



MULTIMEDIA UNIVERSITY OF KENYA

FACULTY OF COMPUTING & INFORMATION TECHNOLOGY

SMART DRINK AND MEAL RECOMMENDATION SYSTEM

BY

NAME: MIGIRO ASA JOSHUA

REG. No: CIT-223-057/2022

SUPERVISOR: NICK GACHUI

FEBRUARY 2022

Project Proposal submitted in partial fulfillment of the requirements of Bachelor of
Science in Computer/Information Technology

DECLARATION

I hereby declare that this Project Proposal is my own work and has, to the best of my knowledge, not been submitted to any other institution of higher learning.

Student : _____

Registration Number:

Signature:

Date:.....

This Project Proposal has been submitted as a partial fulfillment of requirements for the Bachelor of Science in Computer/Information Technology of Multimedia University of Kenya with my approval as the University supervisor.

Supervisor: _____

Signature:

Date:

.....

ABSTRACT

This project presents the design and implementation of a Smart Drink and Meal Recommendation System for a campus environment. The system aims to assist users in selecting suitable meals and beverages by providing personalized recommendations based on user preferences, historical interactions, and contextual factors such as time of day. Many campus cafeterias currently rely on static menus that do not adapt to individual user needs, leading to repetitive choices and reduced user satisfaction. This project adopts an incremental approach, beginning with a rule-based recommendation engine and allowing for future enhancement using machine learning techniques. The system is implemented using a web-based architecture, a relational database, and a simple user interface to demonstrate feasibility and practicality. The results show that context-aware and preference-based recommendations can improve the decision-making process and overall user experience. The project concludes with recommendations for future work, including the integration of advanced learning algorithms and real-time analytics.

Table of Contents

DECLARATION	ii
ABSTRACT.....	iii
LIST OF ABBREVIATIONS.....	vi
CHAPTER ONE: INTRODUCTION.....	1
1.1 Background of the Study.....	1
1.2 Problem Statement.....	1
1.3 Aim of the Study	1
1.4 Significance and Justification of the Study.....	2
1.5 Scope of the Study	2
1.6 Assumptions.....	2
1.7 Limitations.....	2
CHAPTER TWO: LITERATURE REVIEW	3
2.1 Introduction.....	3
2.2.1 Recommendation Systems (RS)	3
2.2.2 Content-Based Filtering (CBF).....	3
2.2.3 Collaborative Filtering (CF).....	3
2.2.4 Context-Aware Recommendation	3
2.3 Related Systems.....	4
2.3.1 Food Delivery Platforms.....	4
2.3.2 Smart Beverage Dispensing Systems	4
2.3.3 E-commerce Recommendation Systems	4
2.4 Limitations of Existing Systems	4
2.5 How Orderly Addresses These Limitations	4
2.6 Conceptual Framework Diagram	5
CHAPTER THREE: METHODOLOGY	6
3.1 Introduction.....	6
3.2 System Development Methodology.....	6
3.2.1 Incremental Development Model.....	6
3.2.2 Algorithm Design.....	6
3.3 Data Collection Methods and Tools.....	7
3.3.1 Data Sources	7
3.3.2 Data Storage.....	7
REFERENCES	8

APPENDICES	8
------------------	---

LIST OF ABBREVIATIONS

The following abbreviations are used in this document:

RS – Recommendation System

CBF – Content-Based Filtering

CF – Collaborative Filtering

ML – Machine Learning

UI – User Interface

DBMS – Database Management System

HCI – Human-Computer Interaction

CHAPTER ONE: INTRODUCTION

1.1 Background of the Study

In recent years, information systems have increasingly been used to personalize user experiences and simplify everyday decision-making. Recommendation systems, in particular, are widely applied in domains such as e-commerce, entertainment, and food services to assist users in selecting items that best match their preferences. Despite this progress, many campus cafeterias and food outlets still rely on static menus that present the same options to all users, regardless of individual tastes or contextual factors such as time of day.

Students and staff within a campus environment make frequent decisions regarding meals and beverages, often under time constraints. The absence of personalized guidance can lead to repetitive choices, dissatisfaction, and inefficient use of available menu options. This project introduces *Orderly*, a smart drink and meal recommendation system branded with the tagline “**Your menu, simplified**”, to address these challenges through preference-based and context-aware recommendations.

1.2 Problem Statement

Most campus food service systems do not provide personalized assistance to users when selecting meals and drinks. Users are required to manually browse menus and make decisions without support from historical preference data or contextual information such as time of consumption. As a result, users may experience decision fatigue, make suboptimal choices, or overlook suitable menu options. There is therefore a need for a system that can analyze user preferences and contextual factors to provide relevant and personalized meal and drink recommendations.

1.3 Aim of the Study

The aim of this project is to design and implement *Orderly*, a Smart Drink and Meal Recommendation System that simplifies menu selection by providing personalized recommendations based on user preferences and time of day.

1.3.1 Research Objectives

The specific objectives of this study are:

1. To design a system that enables users to register, authenticate, and manage their profiles.
2. To develop a database for storing meals, drinks, user preferences, and interaction history.
3. To implement a rule-based recommendation engine for suggesting meals and drinks.

4. To incorporate time-based context (morning, afternoon, and evening) into the recommendation process.
5. To present personalized recommendations to users through a simple and user-friendly interface.
6. To assess the effectiveness of the system based on user interaction and feedback.

1.4 Significance and Justification of the Study

This study is significant as it demonstrates the practical application of recommendation systems within a campus environment. By simplifying meal and drink selection, the system enhances user experience and reduces decision-making effort. Additionally, the project provides an academic foundation for understanding personalization techniques using rule-based logic, with the possibility of extending the system to incorporate machine learning in future work. Food service providers may also benefit from insights into user preferences and consumption patterns.

1.5 Scope of the Study

The scope of this project is limited to the design and implementation of a web-based recommendation system for meals and drinks within a campus setting. The system focuses on personalization using user preferences and time-based context. Features such as online payments, real-time inventory management, and nutritional or medical advice are outside the scope of this study.

1.6 Assumptions

The study assumes that users will provide accurate information regarding their preferences and that the menu data provided by campus food outlets is correct and regularly updated. It is also assumed that users will interact with the system consistently enough to allow meaningful recommendations.

1.7 Limitations

One limitation of the system is the reliance on limited data during initial usage, which may affect the accuracy of recommendations. This limitation is mitigated by adopting an incremental development approach that allows the system to improve as more user interaction data becomes available.

CHAPTER TWO: LITERATURE REVIEW

2.1 Introduction

This chapter reviews existing literature related to Recommendation Systems (RS) and their application in food and beverage services. It discusses foundational concepts, related systems, their limitations, and how the proposed system, Orderly, addresses identified gaps. The review focuses on practical and scalable solutions appropriate for small to medium-sized environments such as campus cafeterias

2.2 Key Concepts

2.2.1 Recommendation Systems (RS)

Recommendation Systems are information filtering systems designed to predict a user's preference for items by analyzing patterns in data (Resnick & Varian, 1997). They are widely used in digital platforms to reduce information overload and enhance decision-making. According to Ricci, Rokach, and Shapira (2015), RS improve user satisfaction by presenting relevant items rather than requiring users to manually search through extensive options.

2.2.2 Content-Based Filtering (CBF)

Content-Based Filtering recommends items based on similarities between item attributes and a user's past preferences. For example, if a user consistently selects sweet beverages, the system recommends other beverages with similar characteristics. CBF is particularly effective in environments with smaller datasets because it does not rely heavily on other users' data (Ricci et al., 2015).

2.2.3 Collaborative Filtering (CF)

Collaborative Filtering recommends items based on similarities between users. If two users share similar tastes, items preferred by one may be recommended to the other (Adomavicius & Tuzhilin, 2005). While CF can achieve high accuracy, it typically requires large volumes of interaction data and may suffer from the cold-start problem, where insufficient data limits recommendation quality.

2.2.4 Context-Aware Recommendation

Traditional RS focus primarily on user-item interactions. However, context-aware systems incorporate additional factors such as time, location, and environment into the recommendation process (Adomavicius & Tuzhilin, 2011). In a campus food environment, time of day is a significant contextual factor influencing meal and beverage selection. Integrating time-based filtering enhances relevance and usability.

2.3 Related Systems

2.3.1 Food Delivery Platforms

Modern food delivery applications implement hybrid RS models combining CBF and CF techniques. These systems analyze order history, ratings, and browsing patterns to generate personalized suggestions. While effective, they depend on large datasets and complex ML infrastructure, which may not be feasible for small campus deployments.

2.3.2 Smart Beverage Dispensing Systems

Smart beverage systems collect data on drink selections and customization trends to analyze user behavior. These systems focus mainly on beverage personalization and trend analytics. However, they typically do not integrate meal recommendations or contextual factors such as time-based filtering.

2.3.3 E-commerce Recommendation Systems

E-commerce platforms use hybrid approaches that integrate CBF, CF, and deep learning models to enhance accuracy (Ricci et al., 2015). Although these systems are highly scalable, their computational complexity and infrastructure requirements make them unsuitable for lightweight campus-based applications.

2.4 Limitations of Existing Systems

Based on the reviewed literature and systems, the following limitations are identified:

Heavy dependence on large datasets for accurate predictions.

High computational and infrastructure requirements.

Cold-start problem in CF-based systems.

Limited contextual adaptation in simplified deployments.

Over-complexity for small-scale environments such as campus cafeterias.

2.5 How Orderly Addresses These Limitations

The proposed system, *Orderly*, adopts a simplified and scalable architecture appropriate for a campus setting. Instead of relying initially on complex ML models, the system employs a rule-based recommendation engine supported by contextual filtering.

Key design improvements include:

- Reduced dependency on large datasets through preference-based rules.
- Lightweight architecture suitable for deployment on standard web infrastructure.
- Explicit integration of time-based context (morning, afternoon, evening).

- Expandable framework that allows future incorporation of ML techniques such as CBF or CF.

This incremental approach ensures feasibility while maintaining alignment with established RS principles.

2.6 Conceptual Framework Diagram

The conceptual framework of Orderly is illustrated below:

User Input (Preferences, Ratings) | v Preference Analysis Module | v Context Filter (Time of Day) | v Recommendation Engine (Rule-Based) | v Suggested Meals & Drinks

This framework demonstrates how user data and contextual factors are processed to generate personalized recommendations.

CHAPTER THREE: METHODOLOGY

3.1 Introduction

This chapter describes the methodology adopted in the design and implementation of *Orderly – Your Menu, Simplified*. It explains the system development approach, data handling procedures, and justification for the selected techniques.

3.2 System Development Methodology

3.2.1 Incremental Development Model

The project adopts the Incremental Development Model (Pressman & Maxim, 2015). This approach allows the system to be developed in small, manageable modules, reducing risk and improving reliability.

The development was divided into the following increments:

Increment 1: User registration and authentication module

Increment 2: Meal and drink database management

Increment 3: Rule-Based Recommendation Engine (RRE)

Increment 4: Context-Aware Filtering Module (CAF)

Increment 5: Testing and refinement

This methodology was selected because:

- It reduces complexity for beginner-level implementation.

- It allows progressive system enhancement.
- It supports future Machine Learning (ML) integration.

3.2.2 Algorithm Design

The core of *Orderly* is a Rule-Based Recommendation Engine (RRE).

Algorithm Flow:

1. Retrieve user preference profile (sweetness, caffeine, meal type).
2. Identify current time category (morning, afternoon, evening).
3. Filter meals by time category.
4. Match drink attributes with user preferences.
5. Rank results based on preference score.
6. Display top recommendations.

Pseudo-code:

```
IF user_pref = sweet
SELECT drinks WHERE sweetness = high
IF time = morning
```

SELECT meals WHERE category = breakfast
RETURN combined ranked results

3.3 Data Collection Methods and Tools

3.3.1 Data Sources

Data used in the system includes: • Predefined meal dataset

- Predefined drink dataset
- User interaction data (ratings and selections)

3.3.2 Data Storage

A Relational Database Management System (RDBMS) is used to store structured data. The system uses Structured Query Language (SQL) for querying and data manipulation.

REFERENCES

Adomavicius, G., & Tuzhilin, A. (2005). Toward the next generation of recommender systems: A survey of the state-of-the-art and possible extensions. *IEEE Transactions on Knowledge and Data Engineering*, 17(6), 734–749.

Adomavicius, G., & Tuzhilin, A. (2011). Context-aware recommender systems. In *Recommender Systems Handbook*. Springer.

Pressman, R. S., & Maxim, B. R. (2015). *Software Engineering: A Practitioner's Approach*. McGraw-Hill Education.

Resnick, P., & Varian, H. R. (1997). Recommender systems. *Communications of the ACM*, 40(3), 56–58.

Ricci, F., Rokach, L., & Shapira, B. (2015). *Recommender Systems Handbook*. Springer.

APPENDICES

Appendix A: User Manual

Appendix B: Project Schedule

Appendix C: Sample Dataset